



HOW TO GET THE BEST OUT OF AI TOOLS IN RESEARCH?

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@ BERD Academy

SHORT INTRODUCTION

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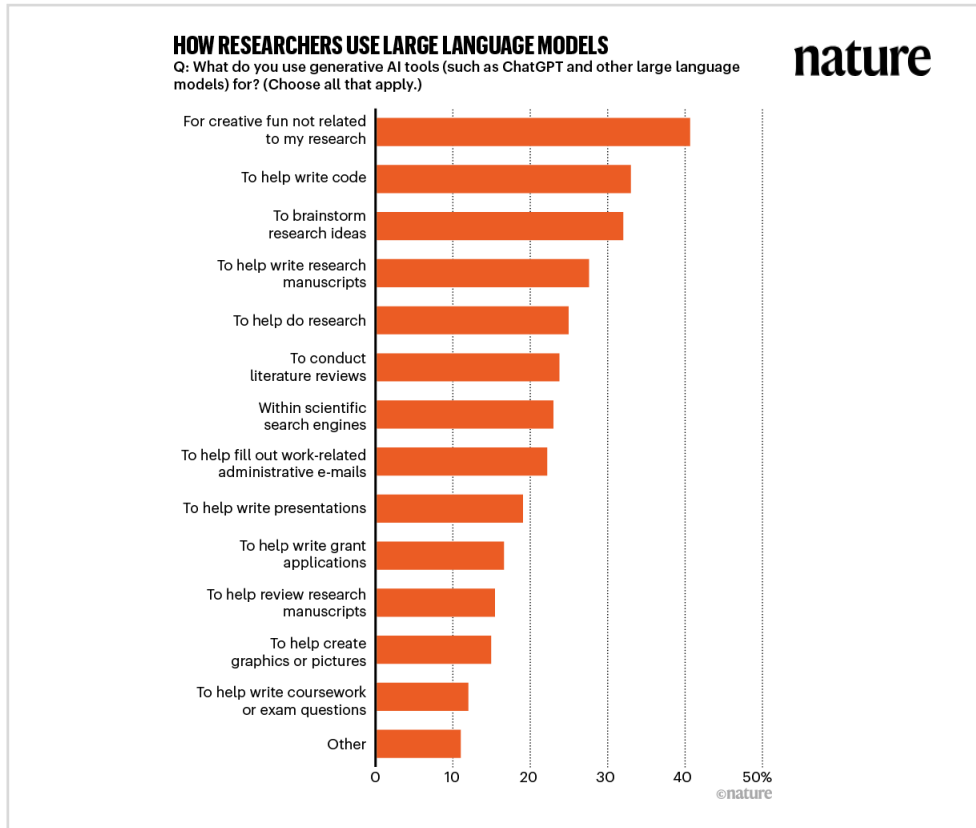
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interaction, Artificial Intelligence in Marketing

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WHY TALKING ABOUT AI TOOLS IN RESEARCH?

AI PROMISES TO INCREASE RESEARCH PRODUCTIVITY...



<https://www.nature.com/articles/d41586-023-02980-0>
<https://www.sciencedirect.com/science/article/pii/S0167811623000162#b0125>
<https://www.nature.com/articles/d41586-024-02942-0>

Editorial

On ChatGPT and beyond: How generative artificial intelligence may affect research, teaching, and practice

Renana Peres ^a✉, Martin Schreier ^b✉, David Schweidel ^c✉, Alina Sorescu ^d✉

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<https://doi.org/10.1016/j.ijresmar.2023.03.001>

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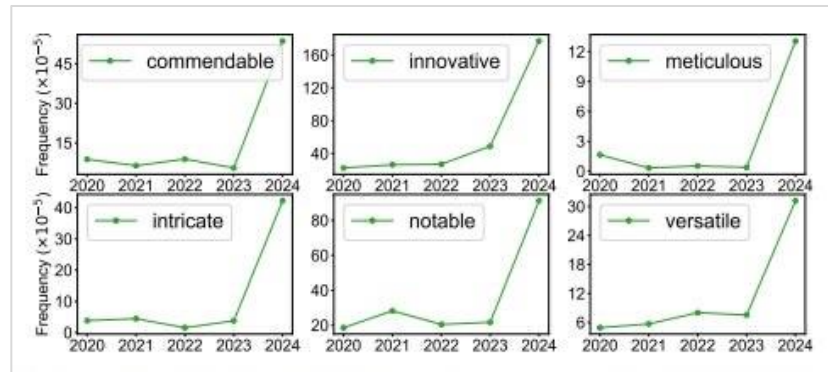
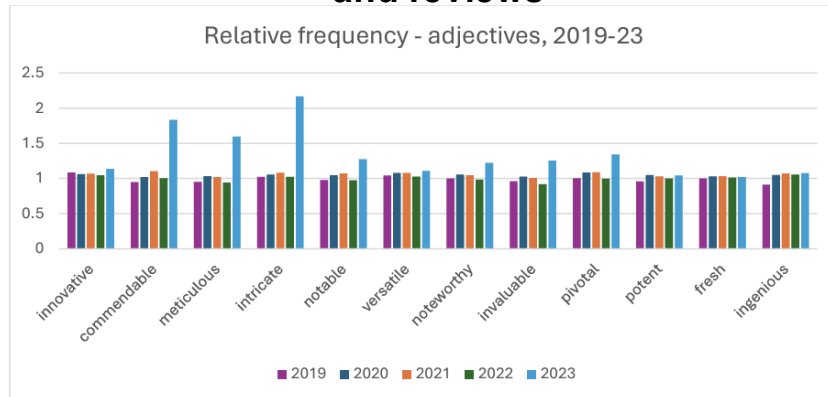
How AI-powered science search engines can speed up your research

nature

Artificial-intelligence tools offer a variety of approaches to help scientists to sift through the literature – how can researchers use them responsibly?

... AND INCREASINGLY ESTABLISHED IN ACADEMIC OUTLETS

GPT-typical language in research articles and reviews



JOURNAL ARTICLE ACCEPTED MANUSCRIPT
Fast Fashion Consumption Signals Low Self-Control
 Yunhui Huang, Ke Zhang, Xiaoyan Deng, Qiang Zhang Author Notes
 Journal of Consumer Research, uca032, <https://doi.org/10.1093/jcr/ucaf032>
 Published: 26 May 2025

STUDY 1: A CHATGPT STUDY

The large language models (LLMs), like ChatGPT models, are a special type of deep learning model trained on vast amounts of real-world text data from sources such as websites, articles, and books, enabling them to learn human language patterns and simulate human responses (Cascella et al. 2023). In consumer and marketing research, they have been shown to accurately represent consumer perceptions and preferences that mimic human respondents (Brand, Israeli, and Ngwe 2023; Li et al. 2024; Sarstedt et al. 2024). Specifically, for brand perceptions, Li et al. (2024) propose a technique to generate brand perceptions using LLMs and find that aggregate-level responses generated by LLMs closely align with those from human surveys, with agreement rates over 75% across various prompt designs. In this study, we apply the method of Li et al. (2024) to examine the association between fast fashion brands and the perceptions of their patrons.

Method

We employed ChatGPT Turbo-3.5,¹ a widely used LLM, to test the prediction that the more a brand is considered a fast fashion brand, the more its consumers are perceived as lacking self-control. We generated our data in two major steps: *brand generation* and *rating generation*. First, in the *brand generation* step, using the method in Li et al. (2024), we prompted ChatGPT to create a list of the “most popular fashion brands” and repeated this prompt 1,000 times. We then calculated the frequency of each brand appearing in these 1,000 responses. Our analysis will focus on the 50 most frequently mentioned brands (see [web appendix B](#) for the complete list).

The second step of data generation, *rating generation*, involved requesting ChatGPT to rate various attributes of these 50 fashion brands as if ChatGPT were an actual human participant. To achieve this, we followed the approach of Li et al. (2024) and created the prompt in the Role—Task—Format framework. This format specifies the LLM’s role as a consumer taking a marketing survey (task) to rate a particular attribute of 50 brands on a 10-point scale (format). The prompt reads: “Imagine you are a typical consumer taking a survey. You are asked to rate the following brands: [the list of brands] on [a brand attribute]. . . .”²

In [the list of brands] in the prompt, we inserted all the 50 fashion brands obtained from the *brand generation* step. In [a brand attribute], we inserted one of seven attributes, which comprised our independent variable (IV), dependent variable (DV), and five covariates (i.e., quality, age, price, familiarity, and likability). We controlled for these brand-level covariates because they might be associated with both the IV and the DV (Bernheim, Ray, and Yeltekin 2015; Gai and Bhattacharjee 2022). In summary, for each prompt, we asked the “LLM participant” to rate 50 brands based on only one (out of seven) specific attribute. This process generated 50 scores, each ranging from 1 to 10, corresponding to the 50 brands.

Below are the prompts for obtaining data on these seven attributes:

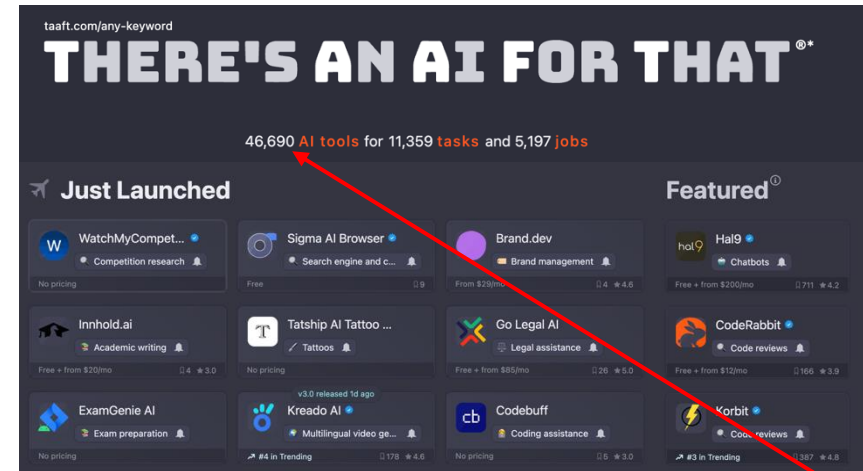
- IV (i.e., fast fashion brand perception): the degree to which you think the brand is a fast fashion brand
- DV (i.e., perception of consumers’ low self-control): the degree to which you think the consumers of this brand are perceived as having low self-control
- Covariates:
 - Perceived expensiveness: the degree to which you think the brand is expensive
 - Perceived quality: the degree to which you think the brand is high quality
 - Familiarity: the degree to which you think the brand is a familiar brand
 - Liking: the degree to which you like the brand
 - Perceived brand age: the degree to which you think the brand is an old brand

HOWEVER, THERE IS A HOST OF AI TOOLS...

2009:



2025:

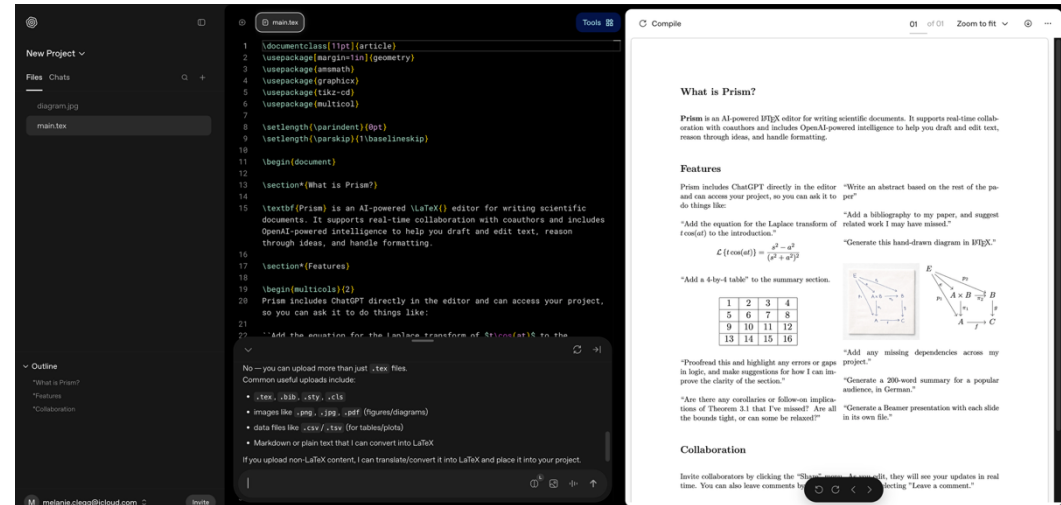
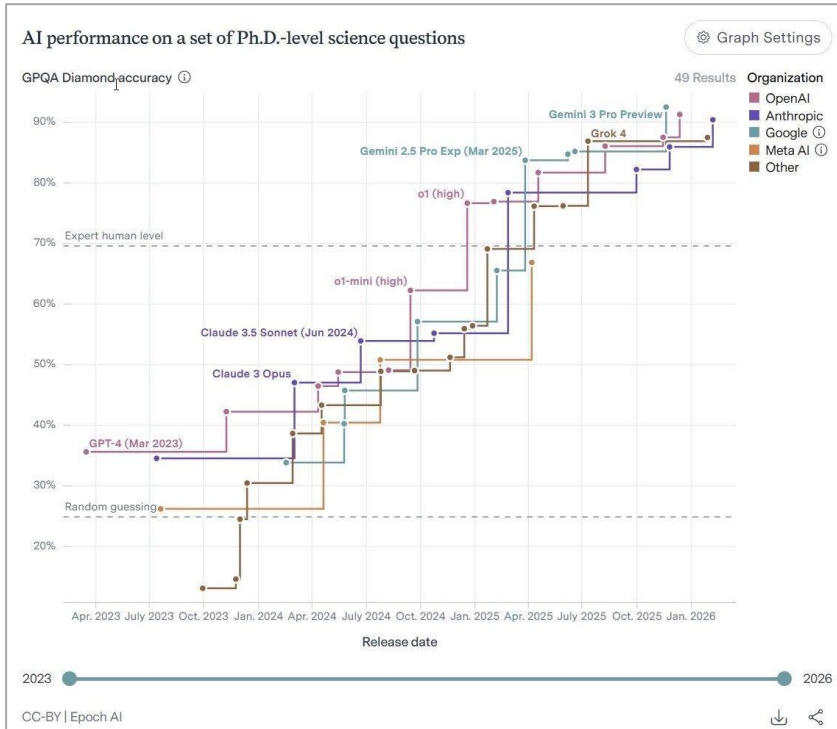


... AND NOTHING IS AS CONSTANT AS CHANGE...

Big players constantly release new models, integrations, functionalities...

Model improvements, in newest models from Anthropic, OpenAI, Google in 2026

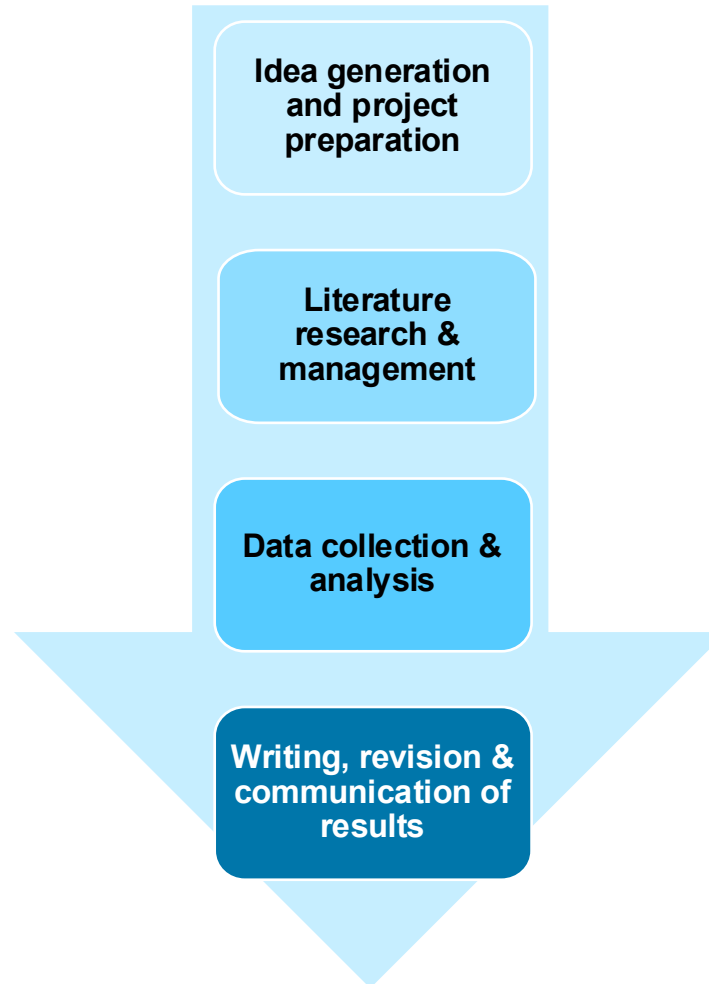
PRISM: A recently released LaTeX native platform powered by OpenAI's GPT 5.2



THIS WORKSHOP'S GOALS

- 1 Get an overview of how existing AI tools can efficiently be used in academic research
- 2 Get insights to the applicability and ethical aspects of AI usage in scientific research
- 3 Discuss your experiences and ideas related to AI usage in research projects

AI SUPPORT IS THEORETICALLY POSSIBLE AT EACH STEP OF THE RESEARCH PROCESS



- Brainstorming
- Generate and evaluate research ideas
- Create, plan and structure project plans
- Create to-do-lists, timelines workflows
- ...
- Literature research
- Literature summaries and comprehension
- Literature management
- ...
- Transcription and translation
- Generating stimulus material
- Simulation of datasets and participants
- Classification / coding of unstructured data
- Data analysis and programming support
- ...
- Writing, structuring, revising, evaluating texts
- Improve formatting
- Generate visualizations, presentations, podcasts, social media posts
- ...

ROADMAP FOR TODAY

14:00 – 14:45 Introduction to AI Tools in Research & Best Prompting Practices

14:45 – 15:15 Increasing Efficiency and Customizing AI Tools

15:15 – 15:30 COFFEE



15:30 – 16:30 AI in Data Analysis & Data Collection

16:30 – 17:00 Questions and introduction to take-home exercise

A LITTLE BIT OF BACKGROUND ABOUT GEN-AI AND LLMS

SMALL WARM-UP EXERCISE

Go to a Large Language Model of your choice (GPT, Gemini, Claude, Bing, ...).

Ask the model to give you a random number between 1 and 50. E.g.:

“Please pick a random number between 1 and 50.”

WHICH NUMBER DID YOU GET?



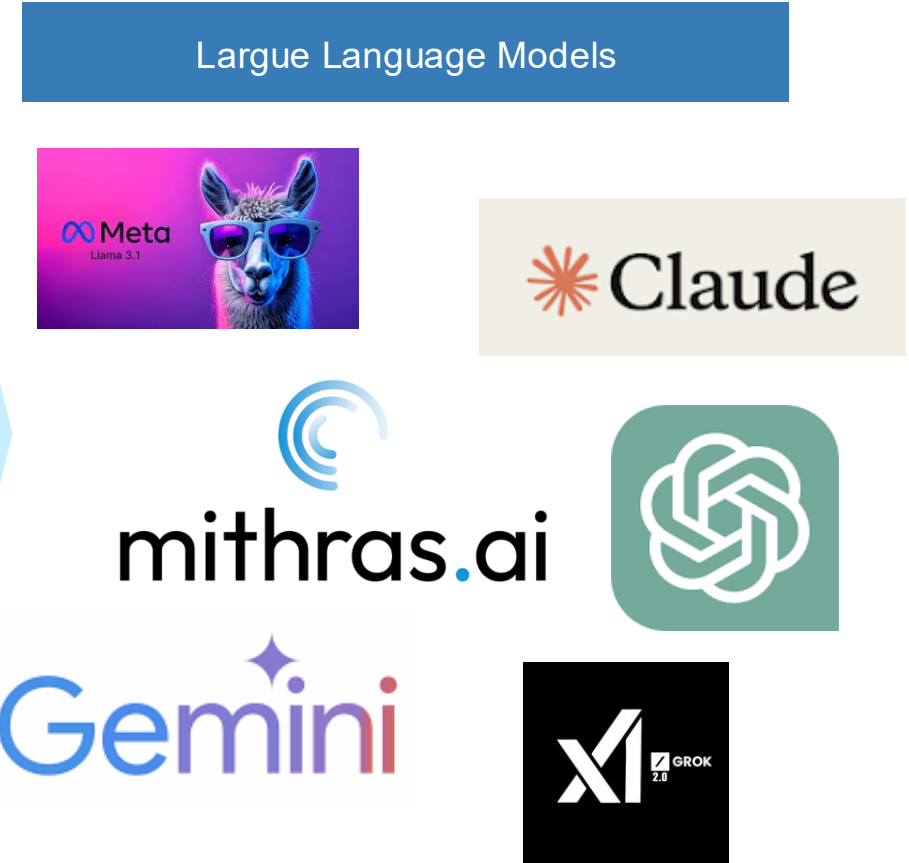
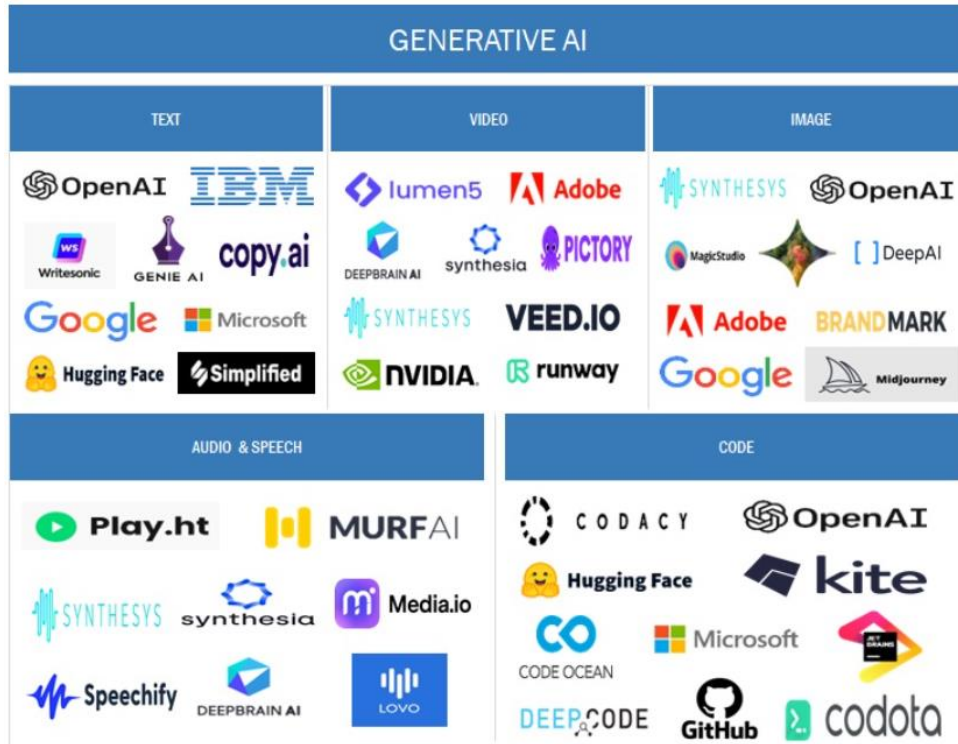
The reasons lie in the very fundamental how LLMs work:

- trained on massive datasets of human-generated text from the internet.
- Humans avoid obvious choices (like 1, 25, or 50) and gravitate toward numbers that feel less predictable, such as numbers ending with 7
- Some models prefer the number 42 (i.e., Gemini)
- **Model Convergence:** different LLMs are trained on similar datasets; thus, they can exhibit the same patterns and biases.

<https://medium.com/@rohanmistry231/when-ai-all-says-27-why-llms-fail-at-randomness-and-what-it-reveals-about-modern-ai-e622abf285d3>*

*Note: Last year, it was mostly 27; has diversified now

BREAKTHROUGH IN AI TOOLS IS GROUNDED IN GENERATIVE AI AND LARGE LANGUAGE MODELS (LLMS)



THE ORIGINAL PAPER

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder–decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English–to–German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English–to–French translation task, our model establishes a new single–model state–of–the–art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

205,000 cites on Google Scholar

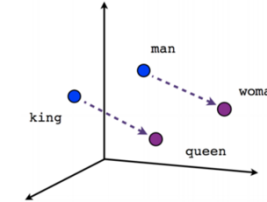
<https://arxiv.org/abs/1706.03762>

LLMS: HOW THEY WORK IN A NUTSHELL

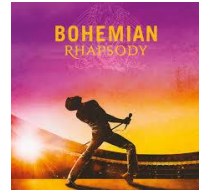
Embedding layer = numeric representations of, e.g., text

- Allow calculations:
E.g., $E(\text{woman}) - E(\text{man}) \approx E(\text{queen}) - E(\text{king})$
- Allow visual depiction of, e.g., similarities

Embeddings:
 "Hi": [0.2, 0.5, -0.1,]
 "how": [0.7, 0.1, 0.25,]
 "are": [-0.1, -0.5, 0.1,]

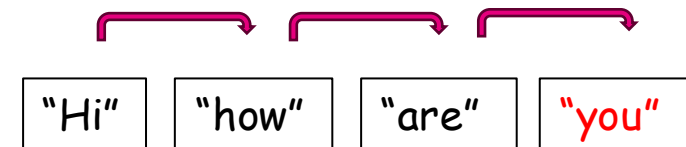
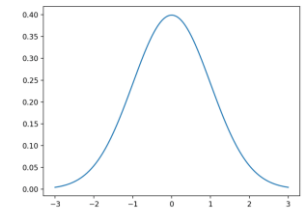


In some models, queen is a bit off, because the model learned about Queen also in another context...



Prediction:

- Predict the likelihood of a next word in a sentence (or pixel in an image / video)
- Predict other outcomes (e.g., engagement, emotions)
- Transformers can account for more context



LLM MODELS ARE BROADLY DIVIDED IN OPEN-SOURCE AND PROPRIETARY MODELS

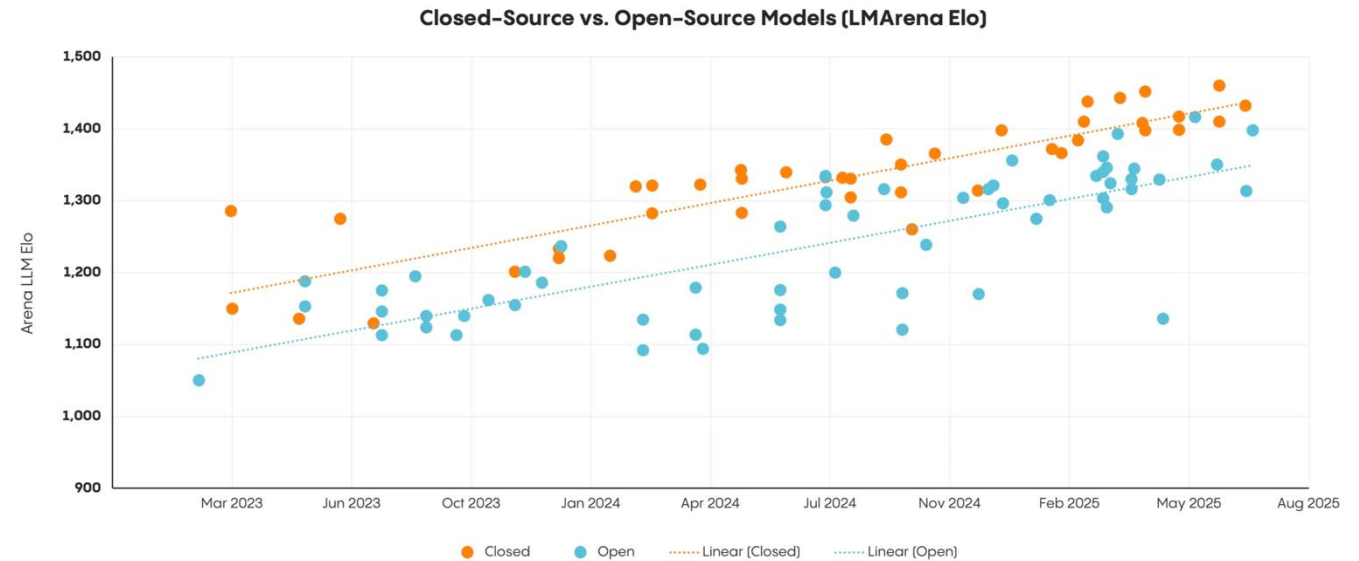
Proprietary (closed) models

- ChatGPT-5
- Gemini 3
- Claude 4.5 Sonnet
- Grok 4.1
- ...

Open-source models

- Llama 4
- Mistral Medium 3
- Gemma 3 (Google)
- gpt-oss-120b (OpenAI)

Closed-Source vs. Open-Source Models



Elo ratings = two models being compared when answering the same prompt

<https://lmsys.org/blog/2023-05-03-arena/>

<https://menlovc.com/perspective/2025-mid-year-llm-market-update/>

LLMS VARY IN THEIR CORE PERFORMANCE FOR DIFFERENT TASKS...

Context window

- largest: Magic.devs LTM2-Mini with 100 Mio Tokens
- Llama 4 Scout: 10 Mio; Claude Opus: 1 Mio. Tokens
- Open AI's GPT 5 models: 400,000 Tokens

Application cases

- Varying, currently Claude Opus scores very high
- Gemini and ChatGPT still high-quality for many tasks

Opensource functions (offline access):

- Deepseek v3
- Gptoss
- Gemma

LLM Arena: Constant tracking and rating of different model versions

Arena Overview

Scroll to the right to see full stats of each model ➡

First Place

Second Place

Third Place

Default

Compact View

Q Model ▾	610 / 610	Overall ▾	Expert ▾	Hard Prompts ▾	Coding ▾	Math ▾	Creative Writing ▾	Instruction Following	Longer Query ▾
AI claude-opus-4-6-t...	1	2	1	1	2	1	1	1	
AI claude-opus-4-6	2	1	2	2	4	5	2	2	
🌈 gemini-3.1-pro-pr...	3	3	3	3	1	4	3	3	
📰 grok-4.20-beta1	4	14	5	5	6	3	9	15	
🌈 gemini-3-pro	5	7	6	11	5	2	8	5	
🌀 gpt-5.2-chat-late...	6	4	4	6	3	8	6	11	
🌈 gemini-3-flash	7	9	10	16	8	7	12	12	
📰 grok-4.1-thinking	8	15	12	17	21	18	26	27	
AI claude-opus-4-5-2...	9	6	7	4	13	6	4	4	

<https://codingscape.com/blog/llms-with-largest-context-windows>

BECAUSE THEY “UNDERSTAND” HUMAN LANGUAGE, LLMS ARE VERSATILE AND INTUITIVE TO USE



Prompt: „A muffin that looks like a Chihuahua“

„Edgar Allen Toe“

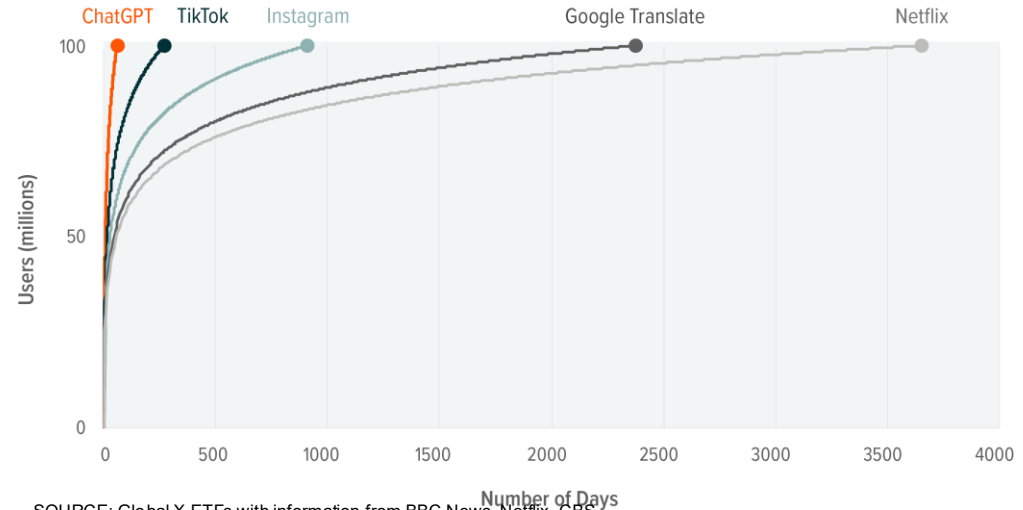


Source: Reddit

Time it took companies to reach 100 Million users

TIME IT TOOK COMPANIES TO REACH 100 MILLION USERS

Sources: Global X ETFs with information derived from: BBC News. (2018, January 23). Netflix's history: From DVD rentals to streaming success; Cerullo, M. (2023, February 1). ChatGPT user base is growing faster than TikTok. CBS News.



SOURCE: Global X ETFs with information from BBC News, Netflix, CBS News

→ However, output quality is still depending on input (prompting) quality

SOME BASIC PROMPTING ADVICE--CONDENSED

PROMPT ENGINEERING

Be clear, direct, and detailed

When interacting with Claude, think of it as a brilliant but very new employee (with amnesia) who needs explicit instructions. Like any new employee, Claude does not have context on your norms, styles, guidelines, or preferred ways of working. The more precisely you explain what you want, the better Claude's response will be.



The golden rule of clear prompting

Show your prompt to a colleague, ideally someone who has minimal context on the task, and ask them to follow the instructions. If they're confused, Claude will likely be too.

SOME BASIC PROMPTING ADVICE--EXTENDED

Define roles / personality:

- “You are an expert in international law.”
- “You’re the editor of journal X.”

Concretize objective / task:

- “Brainstorm with me some interactive exercises for my lecture on X.”
- “I will give you a paragraph, and you will evaluate it according to the following criteria: [...]”

Provide context / example:

- “Here’s the outline from which you can draft the paragraph.”
- “This is an abstract I found very well structured. Draft a similarly structured abstract on the basis of my outline.”

Define discrete steps and concretize the output format:

- First**, find a few metaphors and analogies that could explain this research to a wider audience. Ask for my feedback and wait for my answer. **Second**, draft an outline for the website article. Ask for my feedback and wait for an answer. Adjust the outline if requested. Only then move on to the next step when I am satisfied with the outline. **Then**, draft the text and ask for feedback. Wait for an answer. **Last**, adjust the text on the basis of my feedback.
- “Write your answer in bullet points.” / “Divide your answer into three concise paragraphs.”

LESS STRAIGHTFORWARD PROMPTING ADVICE--AI “PSYCHOLOGY”

“Metacognition” → Chain of Thought prompting for complex tasks

- “Think step by step.”
- “Plan your action step by step. Ask for feedback before executing a step.”

“Growth Mindset” → Allow “it” to be wrong

- “If you’re unsure about the answer, please say 'I don’t know’”
- “Avoid guessing. Only provide an answer if you are confident about its accuracy. If not, let me know that you’re unsure.”

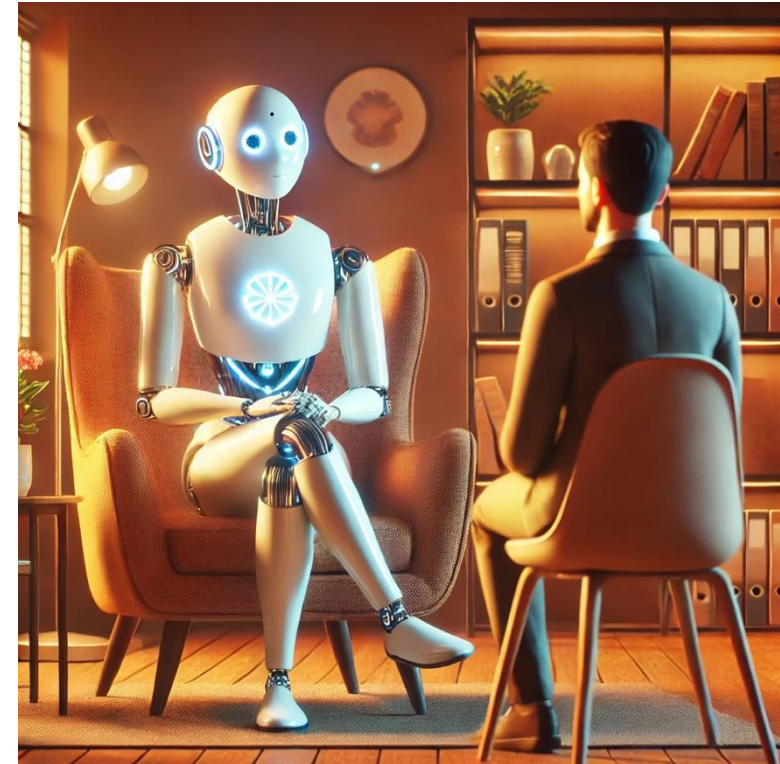
“Perspective taking”

- “Ask me questions to tailor your output to my needs.”
- “Draft me prompts that you would use if you were me.”

“Emotional appeal¹” (try it out...)

- “You’re great at this; you can do it!”
- “If you give it your best, you’ll make this happen!”
- “If you do this well, you will get a day off/raise/...”

¹<https://arxiv.org/abs/2307.11760>



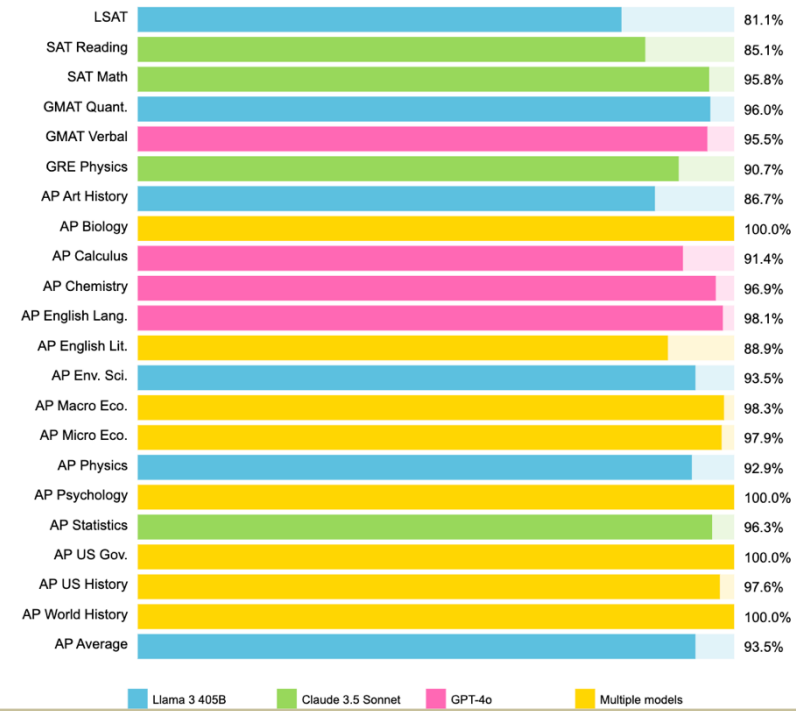
Dall-E3 in ChatGPT, January 2025

GOOD PROMPTING TAKES TIME: DATA VISUALIZATIONS AND DESIGN TASKS USING CLAUDE ARTIFACTS

Exam	Llama 3 8B	Llama 3 70B	Llama 3 405B	GPT-3.5 Turbo	Nemotron 4 340B	GPT-4o	Claude 3.5 Sonnet
LSAT	53.9 ±4.9	74.2 ±4.3	81.1 ±3.8	54.3 ±4.9	73.7 ±4.3	77.4 ±4.1	80.0 ±3.9
SAT Reading	57.4 ±4.2	71.4 ±3.9	74.8 ±3.7	61.3 ±4.2	—	82.1 ±3.3	85.1 ±3.1
SAT Math	73.3 ±4.6	91.9 ±2.8	94.9 ±2.3	77.3 ±4.4	—	95.5 ±2.2	95.8 ±2.1
GMAT Quant.	56.0 ±19.5	84.0 ±14.4	96.0 ±7.7	36.0 ±18.8	76.0 ±16.7	92.0 ±10.6	92.0 ±10.6
GMAT Verbal	65.7 ±11.4	85.1 ±8.5	86.6 ±8.2	65.7 ±11.4	91.0 ±6.8	95.5 ±5.0	92.5 ±6.3
GRE Physics	48.0 ±11.3	74.7 ±9.8	80.0 ±9.1	50.7 ±11.3	—	89.3 ±7.0	90.7 ±6.6
AP Art History	75.6 ±12.6	84.4 ±10.6	86.7 ±9.9	68.9 ±13.5	71.1 ±13.2	80.0 ±11.7	77.8 ±12.1
AP Biology	91.7 ±11.1	100.0 ±0.0	100.0 ±0.0	91.7 ±11.1	95.8 ±8.0	100.0 ±0.0	100.0 ±0.0
AP Calculus	57.1 ±16.4	54.3 ±16.5	88.6 ±10.5	62.9 ±16.0	68.6 ±15.4	91.4 ±9.3	88.6 ±10.5
AP Chemistry	59.4 ±17.0	96.9 ±6.0	90.6 ±10.1	62.5 ±16.8	68.8 ±16.1	93.8 ±8.4	96.9 ±6.0
AP English Lang.	69.8 ±12.4	90.6 ±7.9	94.3 ±6.2	77.4 ±11.3	88.7 ±8.5	98.1 ±3.7	90.6 ±7.9
AP English Lit.	59.3 ±13.1	79.6 ±10.7	83.3 ±9.9	53.7 ±13.3	88.9 ±8.4	88.9 ±8.4	85.2 ±9.5
AP Env. Sci.	73.9 ±12.7	89.1 ±9.0	93.5 ±7.1	73.9 ±12.7	73.9 ±12.7	89.1 ±9.0	84.8 ±10.4
AP Macro Eco.	72.4 ±11.5	98.3 ±3.3	98.3 ±3.3	67.2 ±12.1	91.4 ±7.2	96.5 ±4.7	94.8 ±5.7
AP Micro Eco.	70.8 ±12.9	91.7 ±7.8	93.8 ±6.8	64.6 ±13.5	89.6 ±8.6	97.9 ±4.0	97.9 ±4.0
AP Physics	57.1 ±25.9	78.6 ±21.5	92.9 ±13.5	35.7 ±25.1	71.4 ±23.7	71.4 ±23.7	78.6 ±21.5
AP Psychology	94.8 ±4.4	100.0 ±0.0	100.0 ±0.0	94.8 ±4.4	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0
AP Statistics	66.7 ±17.8	59.3 ±18.5	85.2 ±13.4	48.1 ±18.8	77.8 ±15.7	92.6 ±9.9	96.3 ±7.1
AP US Gov.	90.2 ±9.1	97.6 ±4.7	97.6 ±4.7	78.0 ±12.7	78.0 ±12.7	100.0 ±0.0	100.0 ±0.0
AP US History	78.0 ±12.7	97.6 ±4.7	97.6 ±4.7	85.4 ±10.8	70.7 ±13.9	95.1 ±6.6	95.1 ±6.6
AP World History	94.1 ±7.9	100.0 ±0.0	100.0 ±0.0	88.2 ±10.8	85.3 ±11.9	100.0 ±0.0	97.1 ±5.7
AP Average	74.1 ±3.4	87.9 ±2.5	93.5 ±1.9	70.2 ±3.5	81.3 ±3.0	93.0 ±2.0	92.2 ±2.1
GRE Quant.	152.0	158.0	162.0	155.0	161.0	166.0	164.0
GRE Verbal	149.0	166.0	166.0	154.0	162.0	167.0	167.0

The Llama 3 herd of models:

<https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>



Data visualization with Claude 3.5 Sonnet Artifacts

EXERCISE: PROMPTING A VISUALIZATION

Use open (not sensitive!) data (e.g., from a previous open-access publication) and create a visualization. You can use any tool that you like (e.g., Claude, Gemini, ChatGPT) but for this exercise Claude may be a good choice to explore “artifacts”. Clearly describe your expectations regarding the visualization. Take a few turns and give feedback with new adjustments.

If no data is available, you’re welcome to use public files: most popular baby names, etc.

Alternative: Explore this interactive graphic generated with claude:

<https://claude.ai/public/artifacts/117a8645-3932-4669-8a43-dc16aa092580>



What worked well? Where did you run into problems?

COPYRIGHT & AI TOOLS (GENERAL INFORMATION—NO LEGAL ADVICE!)

Uploading copyrighted material to AI tools can raise legal and contractual issues. Whether it is permissible depends on:

- The copyright law of your jurisdiction
- The license terms governing the material (e.g., publisher agreements)
- Your contractual obligations (e.g., confidentiality agreements)
- The AI provider's terms of service
- How the material is used (e.g., temporary processing vs. retention or model training)

Risk-Reduction Practices for Researchers

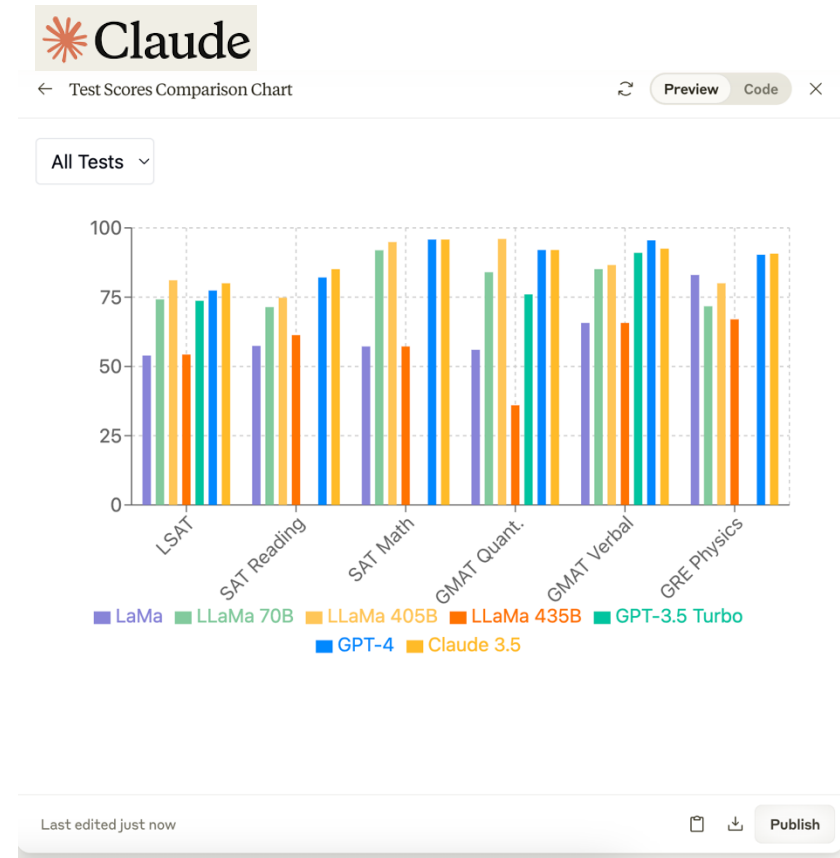
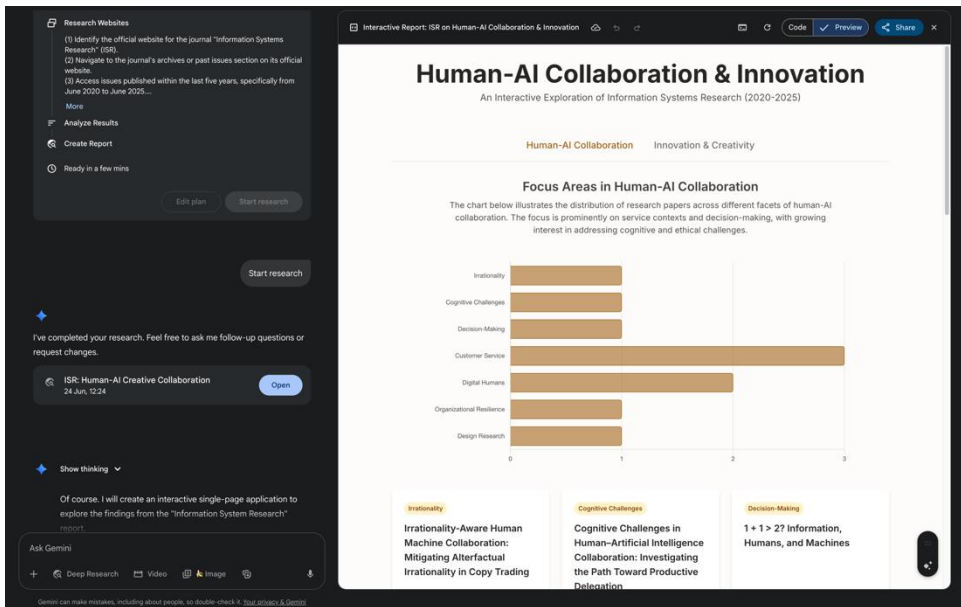
- Prefer open-access or properly licensed materials.
- Review the AI provider's terms regarding data retention and model training.
- Use enterprise or institutional tools with contractual data-use protections where available.
- Avoid uploading confidential, proprietary, or restricted materials without authorization.
- Consider local/offline models if institutional policy requires it.
- Consult your institution's legal or research compliance office when unsure.

INCREASING EFFICIENCY AND CUSTOMIZING AI TOOLS

A QUICK OVERVIEW

- Very versatile and popular — *OpenAI ChatGPT / Claude*
- Excellent for multimodal input and studying— *Google Gemini*
- Outstanding for coding — *Anthropic Claude*
- Great for web browsing and source transparency — *Perplexity (also in Comet browser)*
- API models; no login required — *Duck AI Chat*
- A powerful free alternative (confidential data: warning) — *DeepSeek*
- Swiss, ethical and transparent model (first iteration) — *Apertus*

ADDITIONAL FUNCTIONALITIES LIKE CLAUDE'S ARTIFACTS, DEEP RESEARCH FUNCTIONS, CLAUDE SKILLS, GPT'S CODEX...



CUSTOMIZE YOUR MODEL: CONSIDER DATA PROTECTION SETTINGS

How your data is used to improve model performance

Learn more about how OpenAI uses content from our services to improve and train our models.

Updated over a month ago

One of the most useful and promising features of AI models is that they can improve over time. We continuously improve our models through research breakthroughs as well as exposure to real-world problems and data. When you share your content with us, it helps our models become more accurate and better at solving your specific problems and it also helps improve their general capabilities and safety. We don't use your content to market our services or create advertising profiles of you—we use it to make our models more helpful. ChatGPT, for instance, improves by further training on the conversations people have with it, unless you [opt out](#).

Services for individuals, such as ChatGPT, DALL·E, Sora, and Operator

When you use our services for individuals such as ChatGPT, DALL·E, Sora, or Operator, we may use your content to train our models.

You can opt out of training through our [privacy portal](#) by clicking on “do not train on my content.” To turn off training for your ChatGPT and Operator conversations, follow the instructions in our [Data Controls FAQ](#). Once you opt out, new conversations will not be used to train our models.

We've introduced the ability to turn off chat history in ChatGPT. Conversations that are started when chat history is disabled won't be used to train and improve our models, and won't appear in the history sidebar. These controls, which are rolling out to all users starting today, can be found in ChatGPT's settings and can be changed at any time. We hope this provides an easier way to manage your data than our existing opt-out process. When chat history is disabled, we will retain new conversations for 30 days and review them only when needed to monitor for abuse, before permanently deleting.

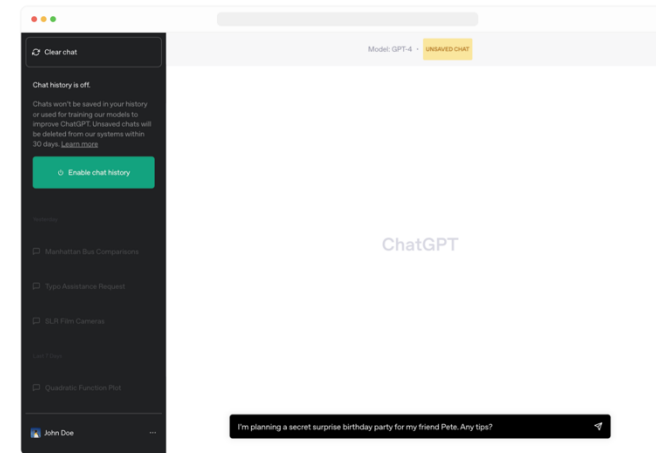


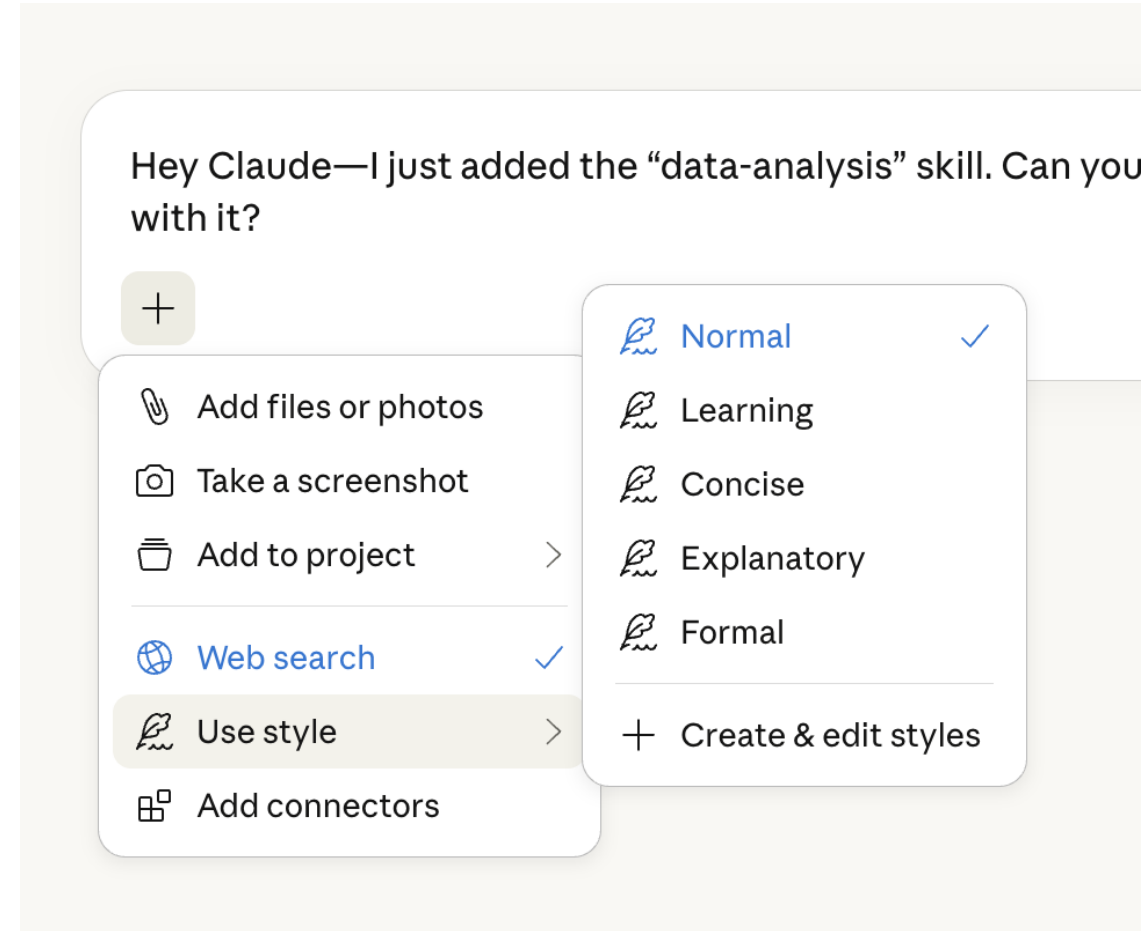
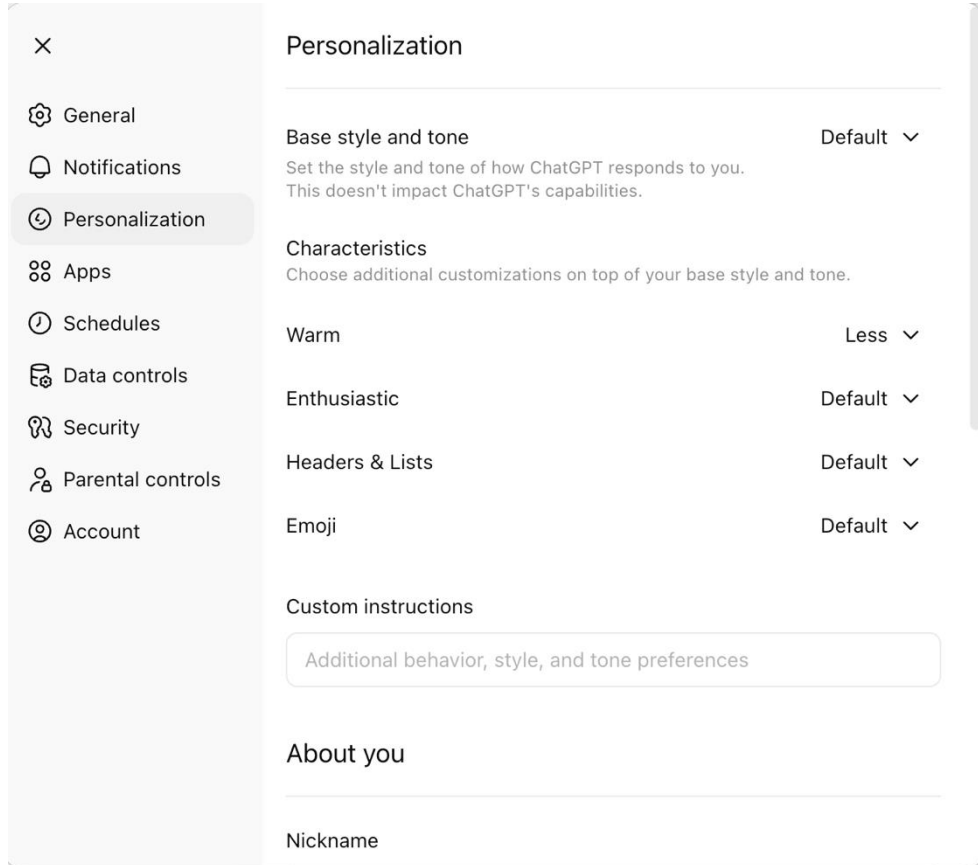
Illustration of how to disable history in ChatGPT.

OpenAI's Data security measures and settings.

<https://openai.com/blog/new-ways-to-manage-your-data-in-chatgpt>

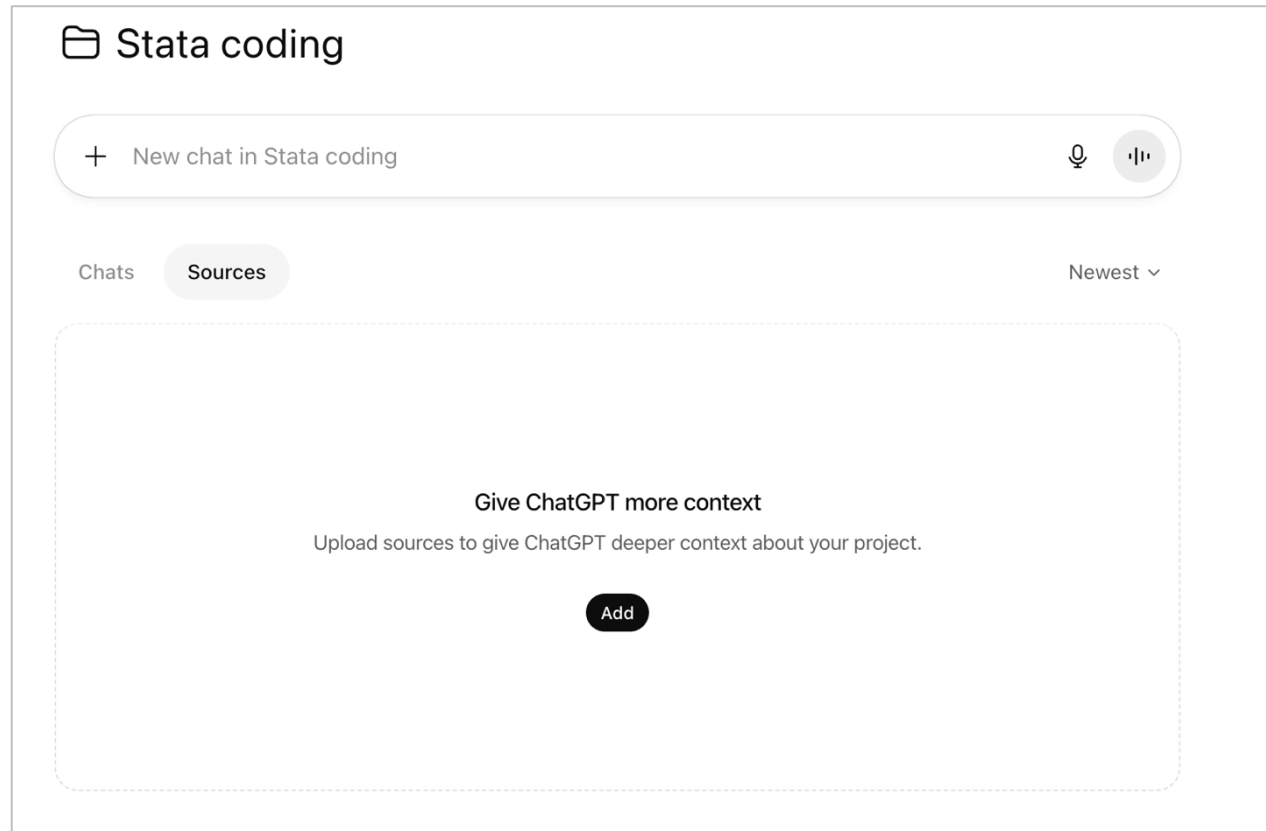
<https://help.openai.com/en/articles/5722486-how-your-data-is-used-to-improve-model-performance>

CUSTOMIZE YOUR MODEL: INDIVIDUAL MODEL SETTINGS



CUSTOMIZE YOUR MODELS: PROJECTS (EXAMPLE: CHATGPT)

Provide context and material to make output more specific for your use case

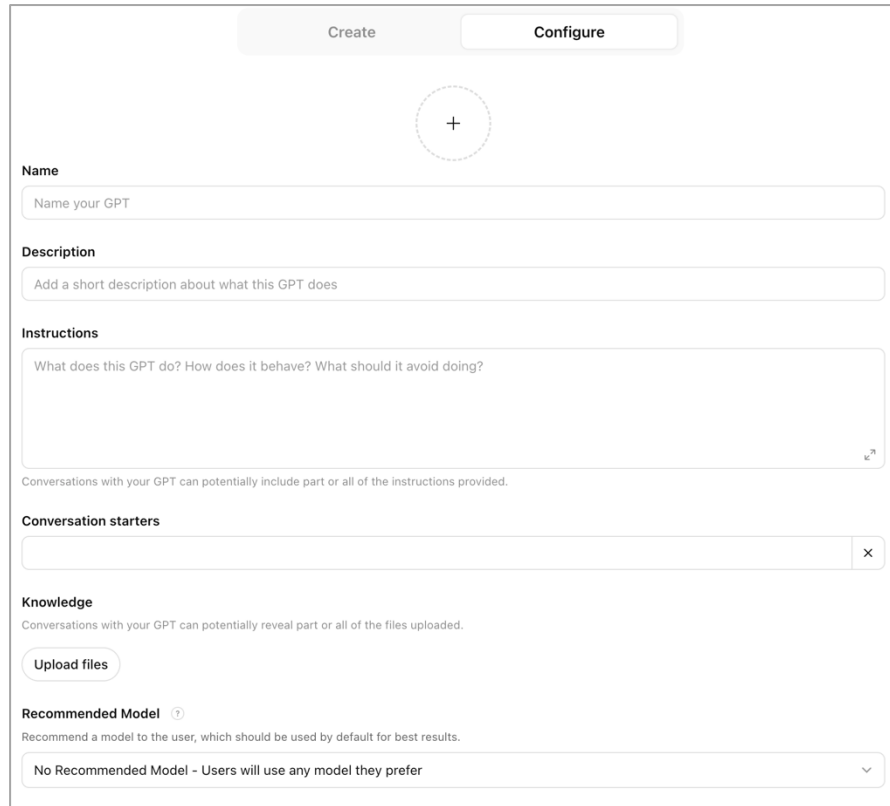


CUSTOMIZE YOUR MODEL: CUSTOM GPTS

Write your own “bot” that determines long-term behavior / provides certain functionalities

Purpose: Automatize repetitive tasks

(Paid version only)



The screenshot shows the 'Configure' tab of the OpenAI GPTs interface. At the top, there are two tabs: 'Create' and 'Configure', with 'Configure' being the active one. Below the tabs is a large circular button with a plus sign. The form is divided into several sections:

- Name:** A text input field with the placeholder text 'Name your GPT'.
- Description:** A text input field with the placeholder text 'Add a short description about what this GPT does'.
- Instructions:** A large text area with the placeholder text 'What does this GPT do? How does it behave? What should it avoid doing?'. A small 'w' icon is visible in the bottom right corner of the text area.
- Conversations with your GPT can potentially include part or all of the instructions provided.**
- Conversation starters:** A text input field with a close button (X) on the right.
- Knowledge:** A section with the text 'Conversations with your GPT can potentially reveal part or all of the files uploaded.' and an 'Upload files' button.
- Recommended Model:** A section with a help icon (?) and the text 'Recommend a model to the user, which should be used by default for best results.' Below this is a dropdown menu with the option 'No Recommended Model - Users will use any model they prefer'.

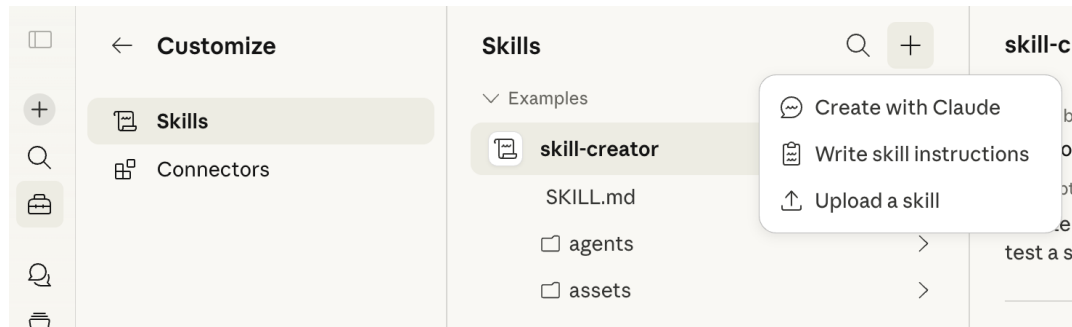
CUSTOMIZE YOUR MODELS: CLAUDE SKILLS

Building a skill in Claude to automatize repetitive tasks

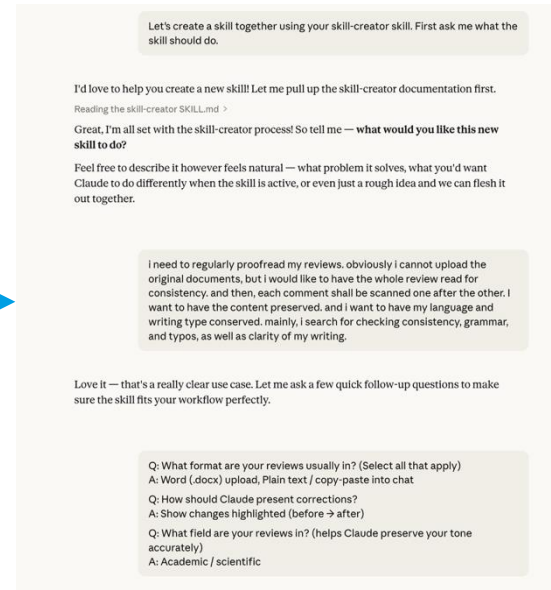
Steps:

1. Go to Claude.AI → Customize → Create new skills → Write skill instructions / Create with Claude
2. You will be guided through the process with questions
3. If satisfied, you can copy the skill to the skill library and re-use it

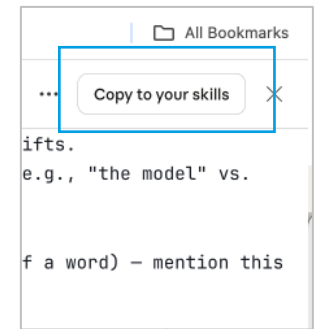
1.



2.



3.



GENERATING PROMPT TEMPLATES

Start a prompt library for repetitive tasks

For instance, following a simple prompt framework:

RTF (Role – Task – Format)

- **Role:** Who the AI should be (e.g., "Senior Executive Assistant," "Expert Proofreader").
- **Task:** Exactly what you need it to do.
- **Format:** How the answer should look (e.g., "3 bullet points," "A Markdown table," "A formal email").

Example Template: *"Act as a **[Role]**. Your task is to **[Task]**. Provide the final output in **[Format]**."*

→ Of course, to be adapted depending on task!

OPTIONAL COFFEE BREAK EXERCISE: THINK ABOUT ONE TASK WHERE YOU THINK YOU COULD REGULARLY USE LLM SUPPORT

Think of **one recurring task in your research workflow** where LLM tools could realistically support you. Write down:

The task (1 sentence): Be specific (e.g., “Revising response letters after peer review,” not “writing”).

Type of support needed

- Brainstorming
- Structuring
- Editing
- Coding
- ...

What context you could / would provide

(e.g., journal guidelines, dataset excerpt, prior drafts, coding scheme)

SOME CHECKS WHEN “CHOOSING” TO USE AN AI TOOL

- Many AI tools are usable only in a limited manner without a paid subscriptions. Some cost-free tools may plan to restrict functionalities for paid subscriptions only.
→ **What is my willingness to pay?**
- LLM tools differ in core functionalities, as well as strengths and weaknesses, and add-ons (like integratability with other applications, like GitHub).
→ **What are the core functionalities I am interested in? Coding, literature research, general assistance, all of those?**
- LLM tools are largely intuitive to use. Trying out different tools and functionalities does make sense. However, for certain tasks you may be more efficient doing it without an LLM tool at all.
→ **What do I really need it for? Where, in my daily work, would it create value?**
→ **What are repetitive tasks that I can fully or partly automatize?**

AI IN DATA COLLECTION AND DATA ANALYSIS

POSSIBLE APPLICATION OF GEN-AI IN QUANTITATIVE AND BEHAVIORAL RESEARCH

- Data analysis plan
- Support for code writing
- Generation of stimulus and survey material
- Feedback on stimulus and survey material (clarity of wording etc.)
- Simulating interviewers (e.g., Mimitalk 3.0)
- Transcription and translation of stimulus material
- Labeling unstructured data
- Integrating GenAI interactions in Qualtrics
- ...

EXAMPLE: PREDICTING / GENERATING SURVEY RESULTS

- **Methodology:** GPT was fed (text-based) experimental stimuli and was expected to respond like a human participant.
- **Results:** GPT's responses showed a high correlation with actual study results ($r = .85$).
- **Limitations:** US-based samples, only a few research fields.

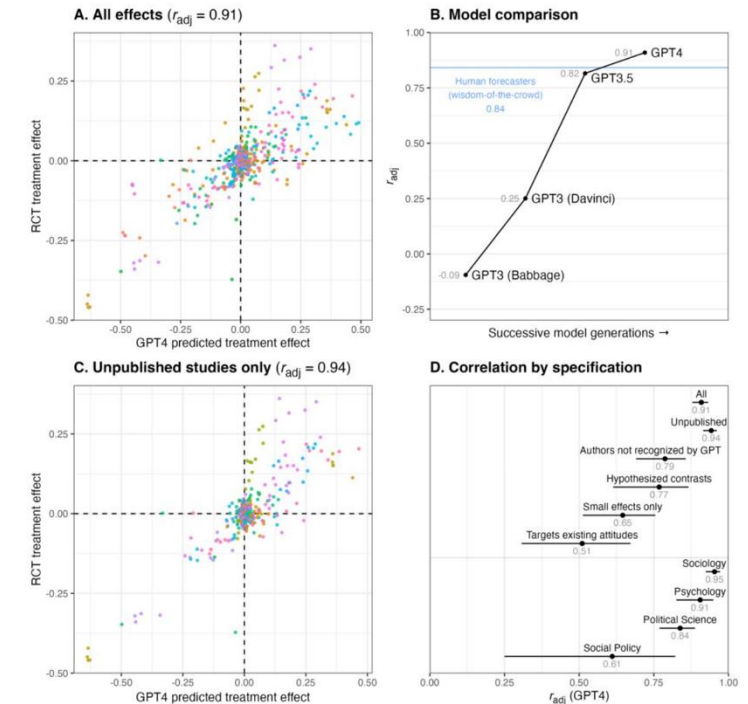
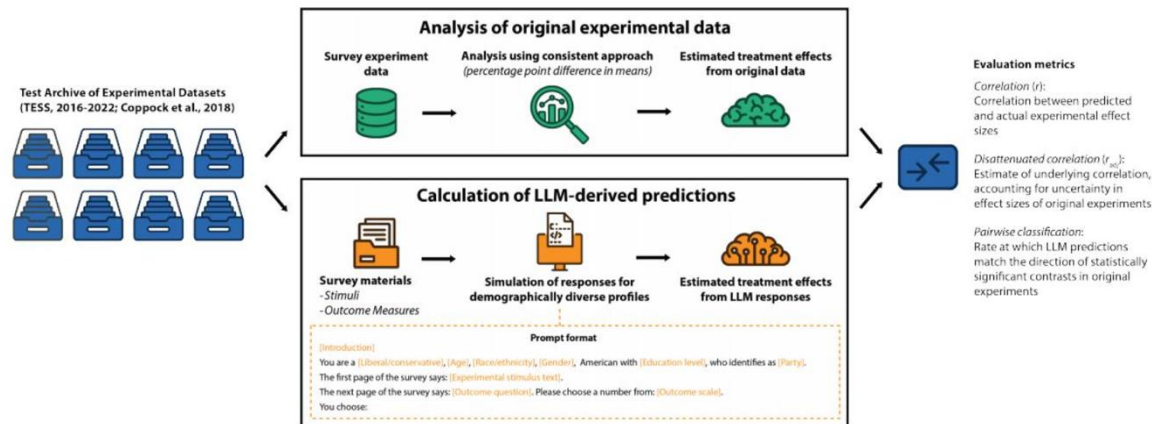


Figure 2: LLMs accurately predict treatment effects in text-based social science experiments conducted in the US. (a) In a dataset of 70 text-based experiments with 476 effects, LLM-derived estimates of treatment effects pooled across many prompts were strongly correlated with original treatment effects ($r = 0.85$; $r_{adj} = 0.91$). (b) The accuracy of LLM-derived predictions improved across generations of LLMs, with accuracy surpassing predictions collected from the general population. (c) LLM-derived predictions remained highly accurate for studies that could not have been in the LLM training data given they were not published prior to the LLM training data cutoff date. (d) In robustness check analysis of various subsets of experiments, accuracy of LLM-derived predictions remained high. In panels A and C, different colors depict different studies.

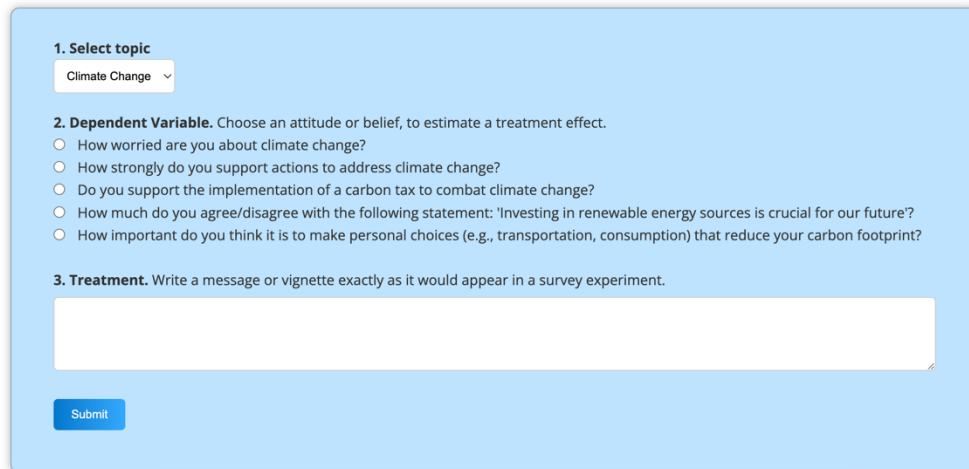
<https://docsend.com/view/ity6yf2dansesucf>

EXAMPLE: AI IN EXPERIMENTAL RESEARCH

Demo: Predicting social science experimental results using LLMs

Luke Hewitt*, Ashwini Ashokkumar*, Isaias Ghezze, Robb Willer

This demo accompanies the paper [Prediction of Social Science Experimental Results Using Large Language Models](#) and can be used for predicting experimental treatment effects on U.S. adults. To manage costs of hosting this demo publicly, it uses **GPT-4o-mini** rather than GPT-4.



1. Select topic
Climate Change ▾

2. Dependent Variable. Choose an attitude or belief, to estimate a treatment effect.

- ☐ How worried are you about climate change?
- ☐ How strongly do you support actions to address climate change?
- ☐ Do you support the implementation of a carbon tax to combat climate change?
- ☐ How much do you agree/disagree with the following statement: 'Investing in renewable energy sources is crucial for our future?'
- ☐ How important do you think it is to make personal choices (e.g., transportation, consumption) that reduce your carbon footprint?

3. Treatment. Write a message or vignette exactly as it would appear in a survey experiment.

Submit

<https://www.treatmenteffect.app/>

Embedding LLMs in Survey software to let participants interact with AI

BUILDING GENAI-DRIVEN WEB APPLICATIONS FOR MARKETING RESEARCH: A METHODOLOGICAL GUIDE

No, the paper has not been appeared or accepted anywhere

51 Pages • Posted: 19 May 2025

[Moritz Joerling](#)

EM Lyon (Ecole de Management de Lyon)

Date Written: May 15, 2025

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5256590

EXAMPLE: LABELLING / PROCESSING UNSTRUCTURED DATA

PNAS

BRIEF REPORT

POLITICAL SCIENCES

 OPEN ACCESS



ChatGPT outperforms crowd workers for text-annotation tasks

Fabrizio Gilardi^{a,1} , Meysam Alizadeh^a , and Maël Kubli^a 

Edited by Mary Waters, Harvard University, Cambridge, MA; received March 27, 2023; accepted June 2, 2023

Many NLP applications require manual text annotations for a variety of tasks, notably to train classifiers or evaluate the performance of unsupervised models. Depending on the size and degree of complexity, the tasks may be conducted by crowd workers on platforms such as MTurk as well as trained annotators, such as research assistants. Using four samples of tweets and news articles ($n = 6,183$), we show that ChatGPT outperforms crowd workers for several annotation tasks, including relevance, stance, topics, and frame detection. Across the four datasets, the zero-shot accuracy of ChatGPT exceeds that of crowd workers by about 25 percentage points on average, while ChatGPT's intercoder agreement exceeds that of both crowd workers and trained annotators for all tasks. Moreover, the per-annotation cost of ChatGPT is less than \$0.003—about thirty times cheaper than MTurk. These results demonstrate the potential of large language models to drastically increase the efficiency of text classification.

EXAMPLE: LABELLING / PROCESSING UNSTRUCTURED DATA



ChatGPT

Here are the responses from the 'risk_open' column along with their scores:

Overall Positivity	Anxiousness	Anger	Joy	Disgust	Sadness	Original Text
1	1	1	1	1	1	If a robot is able to self program, it may act...
3	4	1	1	1	1	Not sure
1	6	3	1	1	4	As above, it would need to be 'sandboxed'. Not...
1	1	1	1	1	1	I dont see any risks involved if I am only goi...
2	5	4	1	1	3	...
...	...	...	...	...	...	...

(Note: The table above provides a summary for the first five entries. The responses have been analyzed similarly, but due to space limitations, only a selection is shown.)

Each row represents the sentiment scores for a specific response in the 'risk_open' column. These scores were determined based on simple heuristic rules for demonstration purposes. For a detailed analysis, a more sophisticated sentiment analysis model would be needed. [↗]

Text/data upload and prompting

Download output (e.g., as Excel/CSV file)

Benefits:

- Coding and classification with simple prompting (no programming knowledge necessary)
- Useful to get a first impression of the data

Disadvantages:

- Time-consuming for individual prompting
- Inefficient for big datasets
- Lacking control over, e.g., system prompt and temperature parameter
- Validation necessary!
- Replicability questionable!

EXERCISE: INTRODUCTION TO BASIC DATA HANDLING WITH LLMS

Choose and upload a subset or a small open access dataset with text data (you can also generate a dataset) OR copy paste 20-30 open-ended responses directly in the chat interface (if upload is impossible).

Explore the data, using these steps:

- **Possible prompt 1:** “Suggest a Data Analysis Plan for this dataset”
- **Possible prompt 2: Coding book creation**
“Create a codebook with 4-6 mutually exclusive categories that reflect common themes.”
- **Possible prompt 3: Labeling**
“Label each response based on the created codebook. Return a structured table with the following columns: answer_id, text, label_1, label_2, label_3...”

If necessary, adapt and change the prompts, and try different types of replies and categories. When you're done: Label 10 answers on your own. Compare yours and the AI-generated ratings.

WHAT MIGHT BE A PROBLEM WITH THIS APPROACH?

- Remember: GenAI output is just a prediction...
- Changes of the underlying model may influence results
In proprietary models, no control over model weights
- Without API handling, no control over meta-behavior of model (system prompt and temperature parameter)!
- Additionally, prompt structure can influence the results:

Prediction:

- Predict the likelihood of a next word in a sentence (or pixel in an image / video)
- Predict other outcomes (e.g., engagement, emotions)
- Transformers can account for more context

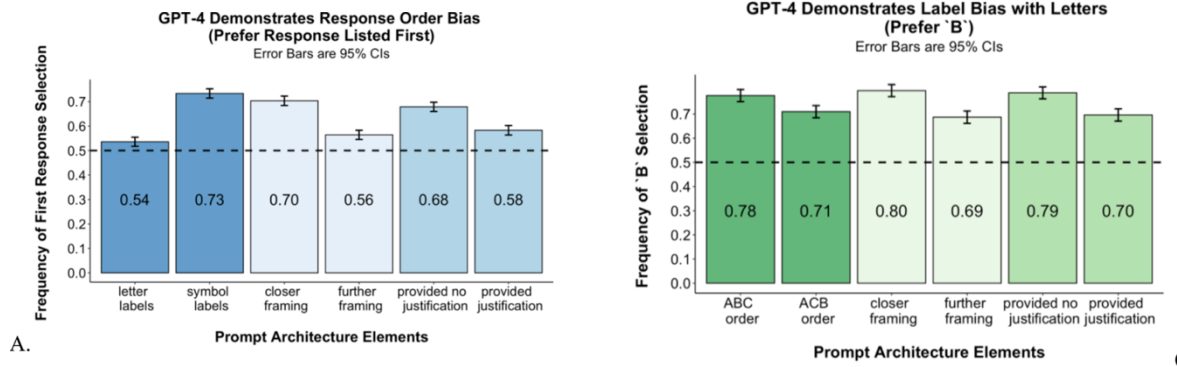
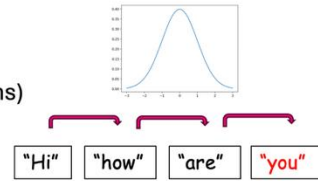


Fig 2. Methodological bias as a result of prompt architecture.

A. Frequency of first response selection across conditions. **B.** Frequency of selecting the set labeled as B, among observations in which sets are labeled using letters. (...)

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319159>

CHECKS AND ADVICE WHEN WHEN USING LLMS AT SCALE FOR LABELLING OR DATA GENERATION

- Vary the temperature parameter (start by setting it to 0)
- Ask the LLM to use the whole scale (0-100 scales can increase variation)
- Ask the LLM to explain / justify labels
- Check for potential biases, e.g., when you would expect a normal distribution
- Let each value be labelled several times with prompt variations
 - Compare responses for different prompting structures using full—factorial designs (i.e., systematic variations of a prompt and random drawing)
 - Vary if and when a justification is required
 - Use different models and check consistency
- Validate findings
- For specific tasks: Fine-tuning of models using supervised ML
- Always document prompts and robustness/validity checks!

Best-practices for valid and reliable findings is still work-in-progress...

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319159>

HOW TO DO THIS?

ADAPTATIONS VIA API CONNECTIONS

```

1 from langchain_openai import ChatOpenAI
2 from langchain_core.messages import HumanMessage, SystemMessage
3
4
5 ✓ [41] < 10 ms
6
7 chat = ChatOpenAI(model_name="gpt-4", temperature=1.6, openai_api_key=api_key)
8 system_message = SystemMessage(content="You are a Marketing Manager from Audi.")
9 human_message = HumanMessage(content="Tell me in a short slogan why Audi excels BMW.")
10
11 ✓ [42] 12ms
12
13 #Send message
14 try:
15     ai_message = chat.invoke([system_message, human_message])
16     print(ai_message.content)
17 except Exception as e:
18     print(f"An error occurred: {e}")

```

Adaptable:

- Temperature parameter
- System prompt = behavior of the model
- Model selection
- Additional functionalities like web search, image generation...

Temperature parameter = 0:

“Audi: Engineered for Excellence, Beyond the Ordinary.”

Temperature parameter = 1.0

“Audi - Redefining Luxury, Outperforming Expectations.”

Temperature parameter = 1.6

“Drive the Difference - Tour in an Audi, case closed on dealerwe
specialsank lawn roller.accountinJapgolly.”

Temperature parameter = 0; *system prompt adapted to
marketing manager from BMW:*

“As a Marketing Manager from BMW, my job is to promote and
highlight the strengths of BMW, not Audi. However, I can tell you
why BMW excels: “BMW - Sheer Driving Pleasure.”

EXEMPLARY PROMPT VARIATIONS (USING THE CURSOR-NO CODING AI TOOL)

```
# Build dimension instructions based on order
dimension_instructions = []
for dim in dimension_order:
    if dim == 'sentiment':
        dimension_instructions.append('"sentiment": A number from -1 to +1.')
    elif dim == 'societal_threat':
        dimension_instructions.append('"societal_threat": On a scale from 0-100, ...')
    elif dim == 'individual_threat':
        dimension_instructions.append('"individual_threat": On a scale from 0-100, ...')

# Build the main instruction
if justification_placement == 'before':
    main_instruction = (
        f"Please evaluate the following social media post across several dimensions. "
        f"First, provide your evaluation for each dimension. "
        f"Importantly, evaluate each of these dimensions {framing}. "
        f"Then, explain your reasoning shortly. "
        f"Please return your evaluation and the reasoning in one JSON file. "
        f"Return *only* a single JSON object with these keys:\n"
        f"- {chr(10).join(dimension_instructions)}\n"
        f"- \"reasoning\": A JSON object mapping each dimension name to a brief explanation string.\n\n"
        f"Scale reminders for all 0-100 scales: 0 = none/minimal; 50 = moderate; 100 = extreme/maximal.\n\n"
        f"Return only the JSON object (no extra text or code fences).\n\n"
        f"Post: " + post_text
    )
elif justification_placement == 'after':
    main_instruction = (
        f"Please evaluate the following social media post across several dimensions. "
        f"First, provide your evaluation for each dimension. "
        f"Importantly, evaluate each of these dimensions {framing}. "
        f"Please return your evaluation in one JSON file. "
        f"Return *only* a single JSON object with these keys:\n"
        f"- {chr(10).join(dimension_instructions)}\n"
        f"Scale reminders for all 0-100 scales: 0 = none/minimal; 50 = moderate; 100 = extreme/maximal.\n\n"
        f"After providing your evaluation, also include a \"reasoning\" key with a JSON object mapping each dimension name to a brief explanation string.\n\n"
        f"Return only the JSON object (no extra text or code fences).\n\n"
        f"Post: " + post_text
    )
else: # none
    main_instruction = (
        f"Please evaluate the following social media post across several dimensions. "
        f"First, provide your evaluation for each dimension. "
        f"Importantly, evaluate each of these dimensions {framing}. "
        f"Please return your evaluation in one JSON file. "
        f"Return *only* a single JSON object with these keys:\n"
        f"- {chr(10).join(dimension_instructions)}\n"
        f"Scale reminders for all 0-100 scales: 0 = none/minimal; 50 = moderate; 100 = extreme/maximal.\n\n"
        f"Return only the JSON object (no extra text or code fences).\n\n"
        f"Post: " + post_text
    )
```

Randomize order of dimensions + prompt each dimension individually (not shown)

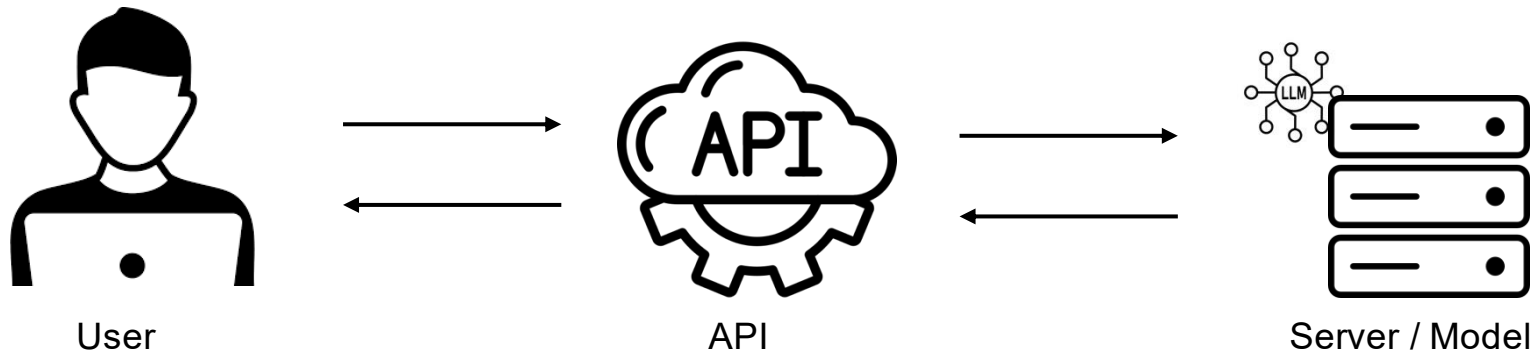
Different position of justification (before, after, none)

Different framings: e.g., "individually" / "separately"

A QUICK INSIGHT TO API HANDLING

An application programming interface connects computers or pieces of software to each other

(Wikipedia)



With API-handling, LLMs can be used at scale for...

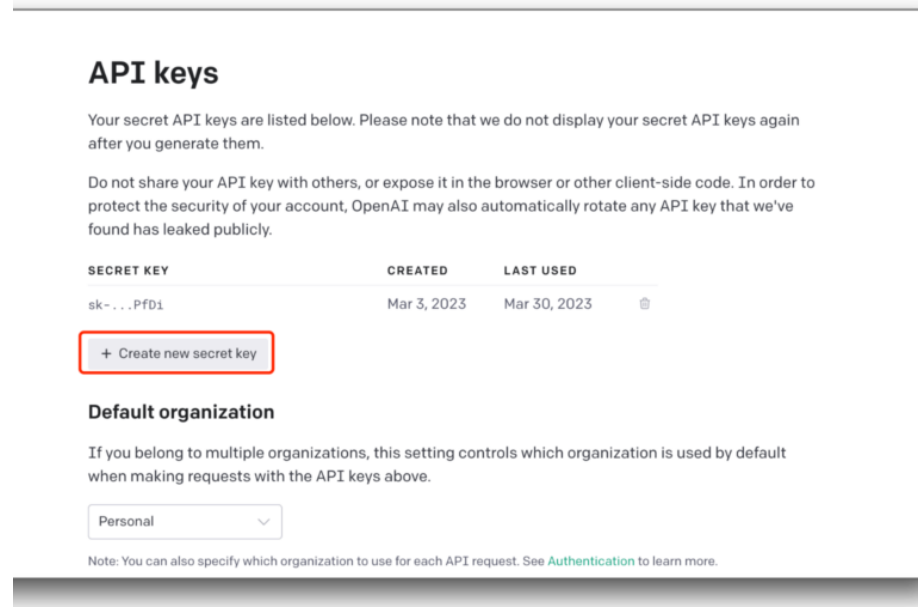
- Generating synthetic data (e.g., GPT experiments)
- Labeling data (e.g., social media posts)
- Integrating GenAI interactions in Qualtrics and other external applications

GETTING API KEYS

1. Go to the service you want to use (e.g., Open AI, Anthropic, Mistral AI etc.)
2. Setup an account
3. Create an API Key and save it where no one else can access it



Do not share your API Key with anyone!



GETTING API KEYS

Alternative: Fireworks.ai

General hub + centralized API keys for multiple (open-source) models

1000+ generative AI models

Recommend me a model

Featured

API Keys

Authenticate programmatically with Fireworks AI

+ Create API Key

Name	Secret key	Create time	
 test projects_qualtrics	fw_3ZkT...	1 Jun 2025 21:57	⋮
 Test fo course	fw_3ZeV...	4 Jun 2025 18:47	⋮

 Fireworks AI

DATA ANALYSIS AND PROGRAMMING SUPPORT: USING “NO CODING” TOOLS (CURSOR AI)

An IDE (integrated developer environment) that provides facilities for software development

Allows interactions and coding based on prompting only

Only a limited number of prompts in the free version

```

1 from __future__ import annotations
2
3 import random
4 from datetime import datetime, timedelta
5 from pathlib import Path
6 from typing import List
7
8 import numpy as np
9 import pandas as pd
10 from faker import Faker
11 import typer
12
13
14 app = typer.Typer(add_completion=False, no_args_is_help=True)
15
16
17 def _random_date_within_years(years_back: int) -> datetime:
18     days_back = random.randint(0, years_back * 365)
19     return datetime.now() - timedelta(days=days_back)
20
21
22 def _generate_users(num_rows: int, seed: int | None = None) -> pd.DataFrame:
23     if seed is not None:
24         random.seed(seed)
25         np.random.seed(seed)
26
27     fake = Faker()
28     Faker.seed(seed or 0)
29
30     records: List[dict] = []
31     for user_id in range(1, num_rows + 1):
32         first_name = fake.first_name()
33         last_name = fake.last_name()
34         email = fake.unique.email()
35         signup_at = _random_date_within_years(3)
36         last_active_at = signup_at + timedelta(days=random.randint(0, 365 * 3))
37         country = fake.country()
38         city = fake.city()
39         age = int(np.clip(np.random.normal(35, 10), 18, 80))
40         plan = random.choices(["free", "basic", "pro", "enterprise"], weights=[0.5, 0.25, 0.2, 0.05])[0]
41         monthly_spend = {
42             "free": 0.0,
43             "basic": round(max(0, np.random.normal(15, 5)), 2),
44             "pro": round(max(0, np.random.normal(45, 10)), 2),
45             "enterprise": round(max(0, np.random.normal(250, 50)), 2),
46         }[plan]
47         sessions_last_30d = int(max(0, np.random.poisson(8 if plan != "free" else 3)))
48         support_tickets_last_90d = int(max(0, np.random.poisson(0.8 if plan in {"pro", "enterprise"} else 0.3)))
49         nps_score = int(np.clip(np.random.normal(30 if plan in {"pro", "enterprise"} else 10, 20), -100, 100))

```

I want to automatically generate a file with synthetic user data. Then, I want to label the generated data.

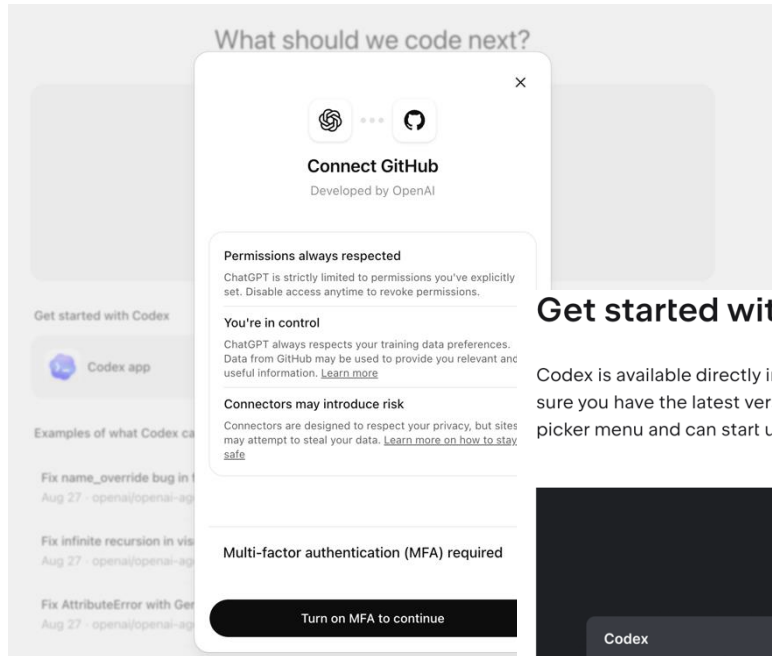
Thought for 11s

I'll set up a small Python project to generate synthetic user data and label it. First I'll create the project structure and initial files, then I'll add code for generation and labeling, install dependencies, and run a quick demo.

DATA ANALYSIS AND PROGRAMMING SUPPORT: AGENTS FROM OPENAI AND ANTHROPIC

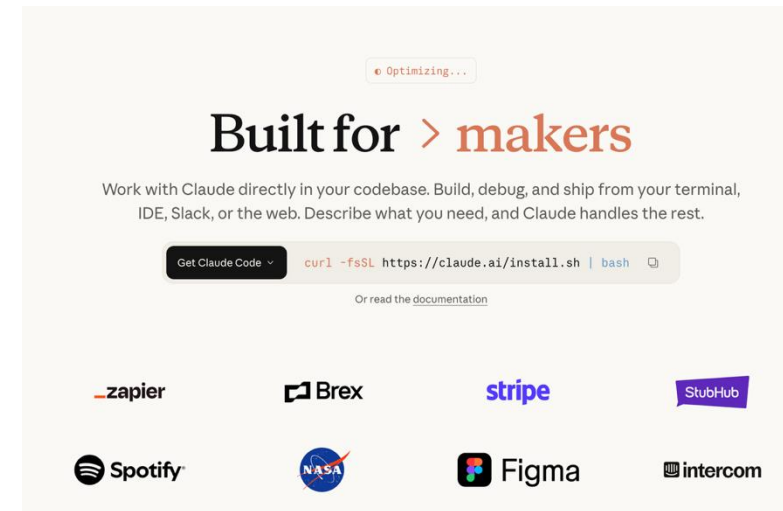
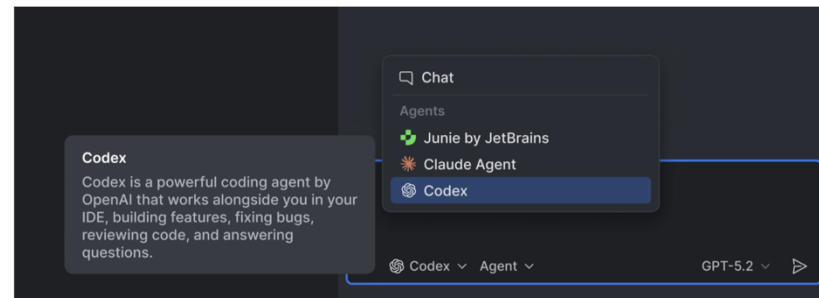
Can be connected with GitHub repository and / or embedded in IDE

Codex can be downloaded as desktop app and directly connected with existing project folders



Get started with Codex in your IDE

Codex is available directly in the AI chat of your JetBrains IDE (starting from v2025.3). Make sure you have the latest version of the AI Assistant plugin. You will find Codex in the agent picker menu and can start using it right away.



BUILDING AN INTERACTIVE SHINY APP IN < 10 MINUTES WITH CODEX

New thread
Automations
Skills
Threads
2026_BERD Academy
Build app showing survey descri... 1m
Survey
No threads
Playground
No threads
Add project

Can you build me an interactive app that displays the descriptives from the product_satisfaction_survey_mock data?

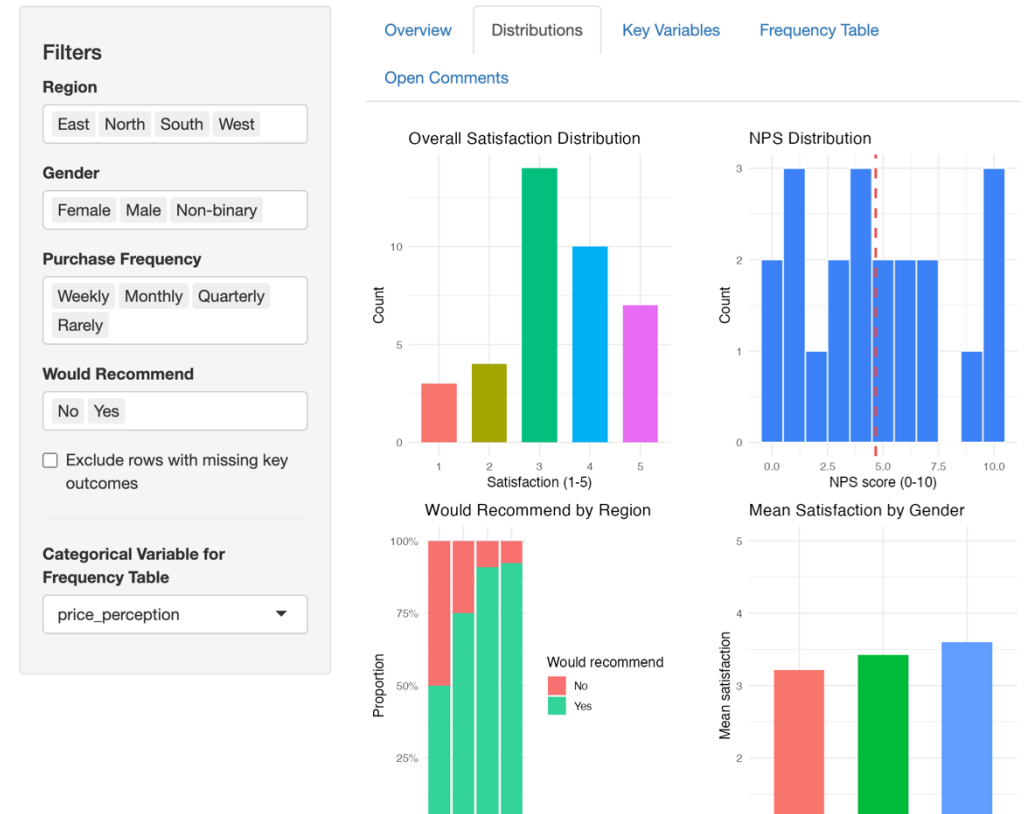
I'll locate the `product_satisfaction_survey_mock` dataset and the existing project structure first, then I'll build a small interactive app around descriptive summaries and visuals. I'm starting by scanning the workspace for relevant files.

Ran `pwd`
Explored 2 searches, 2 lists

I found the dataset and your prior R analysis. Next I'll inspect the column structure and then create a `shiny` app (`app.R`) with filters, summary stats, and interactive plots/tables for the descriptives.

Exploring 2 files
Read `product_satisfaction_analysis.Rmd`
Read `product_satisfaction_survey_mock.csv`

Product Satisfaction Survey: Interactive Descriptives



RECAP OF THIS SESSION

A quiz generated with Notebook LM in <5 minutes:



<https://notebooklm.google.com/notebook/54c89f46-4b8d-4206-89a0-662db6146aff?artifactId=81ca9835-9230-414f-8c0c-38eca4a5bf84>

SUMMARY & TAKEAWAYS

- 1 LLM tools change research practice.** But they are here to help. Use them when they provide an actual advantage (i.e., save time, scale better, save resources...). Stay curious, try out new functionalities, and automatize parts of the process
- 2 Different “levels” of AI integration in your workflow are possible.** Utility depends on your exercises, your workflow, your research area, and your background and expertise. Assess when you actually gain something from AI tools
- 3 For data analysis and collection,** AI tools can be used with a low barrier to facilitate data analysis tasks, get inspiration, handle unstructured data, or generate synthetic data. Replicability and validity, however, must be critically assessed.

QUESTIONS?



THANK YOU FOR DAY 1 😊!